

MUSHAB MAHIN M A

Ernakulam, Kerala

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PROFESSIONAL SUMMARY

AI & Data Science undergraduate with experience building AI-powered backend systems using Python, FastAPI, PostgreSQL, and machine learning. Developed projects in conversational AI, anomaly detection, computer vision, and IoT systems with a focus on scalable APIs and real-time applications.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, R, C

Frameworks/Libraries: FastAPI, TensorFlow, OpenCV, SQLAlchemy

Core Areas: Machine Learning, NLP, JWT Authentication, Computer Vision, LLM, RAG, IoT Systems

Tools/Platforms: Git, GitHub, Docker, PostgreSQL, Postman, Data Visualization

AI/Embedded: Whisper, Grok API, GSM Communication, Sensor Fusion

EXPERIENCE

Frontend Developer Intern

Jan 2023 – Jul 2023

Yallo

Remote

- Developed responsive frontend interfaces using Nuxt.js and Tailwind CSS for cross-device compatibility and improved UI consistency.
- Collaborated with designers and developers to convert UI concepts into reusable production-ready components.
- Optimized frontend responsiveness and resolved browser compatibility issues to improve application stability.

PROJECTS

VOCA AI – Conversational Intelligence Backend | *Python, FastAPI, Whisper, Grok API*

GitHub Repository

- Engineered an API-driven multimodal AI backend to process voice and text conversations using Whisper and LLM-based analysis workflows.
- Implemented speech-to-text transcription, sentiment analysis, intent detection, entity extraction, and escalation-risk identification pipelines.
- Designed scalable FastAPI endpoints and structured JSON response systems for enterprise conversational intelligence integration.

INSIDRA – Insider Threat Detection System | *Python, Flask, TensorFlow, PostgreSQL, React.js*

GitHub Repository

- Built a machine learning-based insider threat detection platform using behavioral drift analysis and dynamic risk scoring techniques.
- Implemented real-time anomaly detection and explainable AI-generated alerts for enterprise security monitoring workflows.
- Integrated backend APIs, database systems, and frontend dashboards into a scalable full-stack monitoring solution.

SmartHelmet AI – Emergency Response System | *ESP32, GPS, GSM, Embedded Systems*

GitHub Repository

- Developed a real-time embedded safety system integrating sensor fusion, GPS localization, and GSM communication for automated accident response.
- Implemented ESP32-based decision logic with fail-safe emergency workflows, helmet-wear detection, and 10-second manual override protection.
- Achieved real-time monitoring performance with 50–100 Hz sensing loops, 80–150 ms event recognition latency, and GPS accuracy within 3–8 meters.

ThirdEye AI – Smart Energy Management System | *Python, YOLOv8, OpenCV, ESP32*

GitHub Repository

- Built an AI-powered energy automation system using YOLOv8-based human detection and IoT-driven device control workflows.
- Integrated ESP32 relay automation and zone-mapping logic for intelligent real-time electrical device management.

EDUCATION

Muthoot Institute of Technology and Science

2023-2027

B.Tech in Artificial Intelligence & Data Science — CGPA: 8.86/10

Kochi, Kerala

LEADERSHIP

Class Representative (Semesters 3–6) – Represented 60+ students as liaison between faculty and students, coordinating academic communication and issue resolution across multiple semesters.